



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS : IX
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 12.12.22

General Instructions:

- The paper consists of three printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1i) A girl walks 200m towards East and then she walks 150m towards North. [8x1 = 8]
The distance of the girl from the starting point is

- a) 350m b) 250m c) 300m d) 225m

ii) A diagonal of a rectangle is inclined to one side of the rectangle at 25° . The acute angle between the diagonals is

- a) 55° b) 50° c) 40° d) 65°

iii) The value of $2\cos^2 30^\circ - 1$ is equal to

- a) $\tan 45^\circ$ b) $\cos 30^\circ$ c) $\cos 60^\circ$ d) $\sin 45^\circ$

iv) What is the value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ$

- a) 1 b) -1 c) $\frac{1}{3}$ d) $\frac{3}{11}$

v) The point which lies on y axis at a distance of 5 units in the negative direction of y axis is

- a) (0,5) b) (5,0) c) (0,-5) d) (-5,0) vi) The distance of the point P(-6,8) from origin is

- a) 8 units b) $2\sqrt{7}$ units c) 10 units d) 6 units

vii) ABC is an Isosceles triangle right angled at C, then which one is correct

- a) $AB^2 = 2AC^2$ b) $AB^2 = 2AC$ c) $AC^2 = 2AB^2$ d) $AC^2 = 2AB$

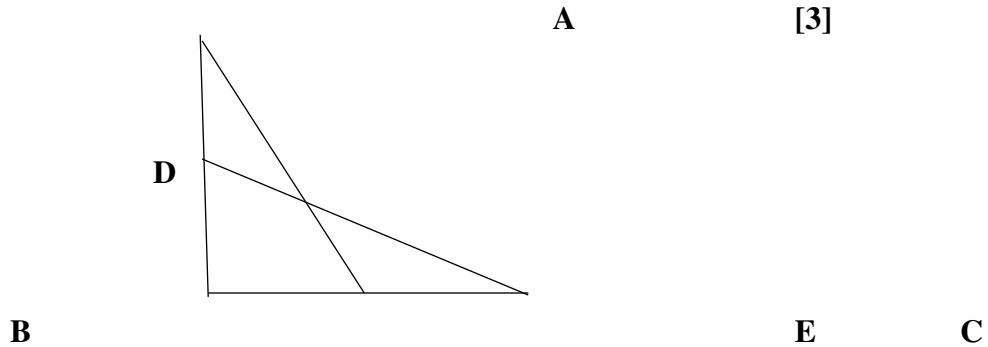
viii) Which of the following is not true for a parallelogram

- a) Opposite sides are equal
b) Opposite angles are equal

- c) Diagonals bisect each other
- d) Opposite angles are bisected by the diagonals.

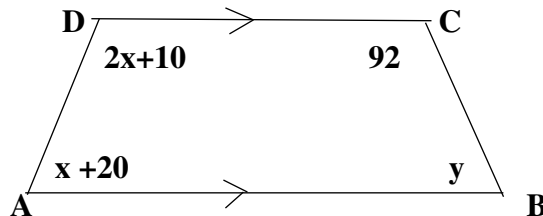
SECTION: B

2. In the given figure, $AE = DC = 13\text{cm}$, $BE = 5\text{cm}$, $\angle ABC = 90^\circ$ and $AD = EC = x\text{ cm}$. Calculate the length of AB and the value of x .



3. In the given figure ABCD is a trapezium. AB is parallel to CD . Find the value of x and y .

[3]



4. Find the value of

$$2\sqrt{2} \cos 45^\circ \cos 60^\circ + 2\sqrt{3} \sin 30^\circ \tan 60^\circ - \sin 27^\circ \sec 63^\circ$$

[3]

5. Find the relation between x and y such that the point (x,y) is equidistant from the points

$(3,6)$ and $(-3,4)$

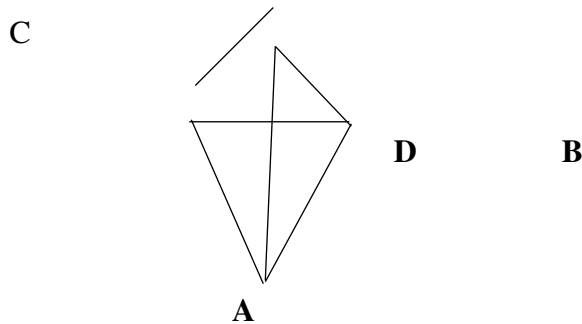
[3]

6. The point $A(0,3)$, $B(-2,a)$ and $C(-1,4)$ are the vertices of a right angled triangle at A , find the value of a .

[4]

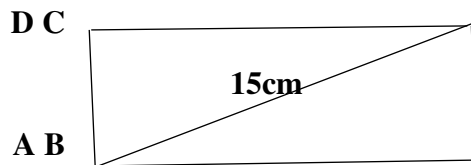
7. In the given figure , ABCD is a kite in which $AB=AD$ and $BC=CD$. Prove that
i) AC is a bisector of angle A and of angle C ii)
AC is the perpendicular bisector of BD.

[4]



8. In the given figure , ABCD is a rectangle . Its diagonal $AC = 15\text{cm}$ and angle $ACD = x^\circ$. If $\cot x^\circ = 3/2$, find the perimeter and area of the rectangle.

[4]



9. Prove that the sum of the squares of the sides of a rhombus is equal to the sum of squares of its diagonals. [4]

10. Show that the points $(1,7)$, $(4,2)$, $(-1,-1)$ and $(-4,4)$ are the vertices of a square.

[4]